Replacing 3D Convolutions with Factorized Spatial Temporal Attention for Soccer Ball Action Spotting

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***Abstract*—Ball action spotting in soccer videos aims to automatically identify and temporally localize events such as passes and ball carries. This task requires modeling both visual information and temporal relationships between video frames. This paper presents a Transformer based approach for the SoccerNet Ball Action Spotting Challenge 2023, investigating the use of factorized spatial temporal attention as an alternative to the 3D convolutional temporal encoder of an existing solution. The proposed architecture uses an ImageNet pretrained EfficientNetV2-B0 backbone to extract spatial features from groups of sampled video frames. These features are projected into a shared embedding space and organized as spatial tokens. A temporal Transformer models the evolution of features independently at each spatial location, followed by a spatial Transformer that captures relationships between locations at each time step. The resulting representations are aggregated to predict the two target event classes: passes and drives. The model is evaluated using the seven fold cross-validation protocol adopted by the original solution. It achieves a mean mAP@1 of 77.45%, with a standard deviation of 5.47 percentage points across folds. This result is 1.61 percentage points below the 79.06% cross validation mAP@1 reported for the original CNN based approach. The findings provide an initial assessment of factorized spatial temporal attention for ball action spotting and highlight its potential as an alternative to convolutional temporal modeling.**

***Index Terms*—Ball action spotting, soccer video analysis, temporal action localization, 3D convolutional neural networks, Transformers, video understanding.**

I. Introduction

The improvements in deep learning and increase computing power permitted considerable progress in the field of sports video understanding. In particular sport video understanding field has become more important area for computer vision researcher [1] [2] [3] moving from easier task like object detection to more demanding challenges like temporal action spotting.

Temporal action spotting is a computer vision task that identifies the exact timestamp when a specific action or event begins or occurs in an untrimmed video, without needing to outline its full start and end duration.

To advance research in soccer video understanding the SoccerNet Challenge was proposed at CVPR 2023. The challenge focused on 7 tasks divided into three categories: Broadcast video understanding, Field Understanding, and Player understanding. We will focus on the first edition of Ball action spotting challenge included in the first category.

A. Cioppa et al. [4] define ball action spotting challenge as localizing the exact time when two types of actions related to the soccer ball occur: pass and drive. This task pose a unique challenge due to the dynamic and complex nature of the game. During a match players move rapidly, often obscuring each other, making it difficult to accurately track their movements. In this challenge a precise timestamp is set for this actions:

- **pass event**: the timestamp correspond to the moment when the ball leaves a player's foot
- **drive event**: the timestamp corrrespond to the moment when a player control the ball

The dataset consist of 7 games with 11,041 annotated timestamps with 2 games not publicy available for the challenge set. The recordings of the games are untrimmed and provided by Footovision. Other details of the challenge can be found at SoccerNet Ball Action Spotting challenge.

The main contributions of this work are:

- We investigate a Transformer based temporal encoder for ball action spotting, replacing the 3D convolutional component of an existing architecture.
- We describe an architecture that combines a shared 2D visual feature extractor with self attention based temporal feature aggregation.

II. Related Work

A. Soccer Action Spotting

Soccer action spotting has received increasing attention in computer vision, particularly with the introduction of the SoccerNet benchmark [1], [4]. The benchmark provides broadcast soccer videos annotated with temporally localized events, enabling the development and evaluation of methods for detecting relevant moments in long, untrimmed recordings. The SoccerNet 2023 challenges further expanded this research direction by introducing several tasks addressing broadcast video, field, and player understanding [4]. Within this framework, Ball Action Spotting focuses specifically on identifying two ball related events: passes and drives. Each event is represented by a timestamp corresponding to a precisely defined interaction with the ball. This formulation differs from broader action spotting tasks because the target events are closely

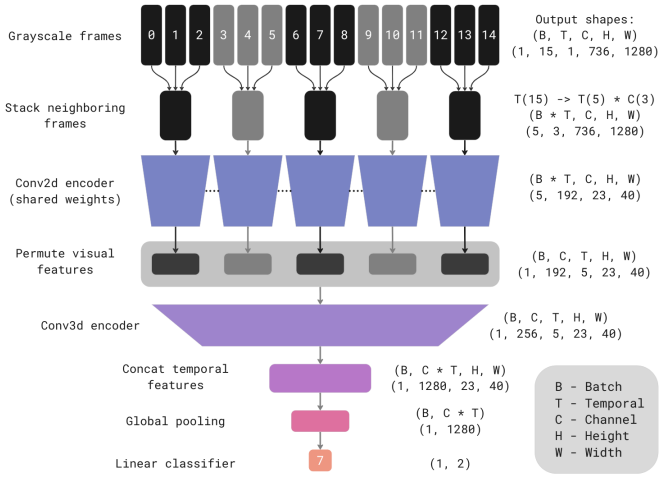


Fig. 1. Architecture of Baikulov's ball action spotting model, from [5]

related to changes in ball possession and control.

The temporal precision required by this task makes the extraction of information from neighboring frames particularly important. A single frame may not provide sufficient context to distinguish the relevant interaction, motivating architectures that combine spatial feature extraction with temporal modeling.

B. Ball Action Spotting

The SoccerNet Ball Action Spotting Challenge 2023 attracted several approaches based on deep learning architectures and different training strategies. Among them, Baikulov achieved first place with an efficient architecture combining 2D and 3D convolutional operations [5].

The proposed solution, reported in Image **??**, adopts a slow fusion strategy. A shared 2D convolutional backbone independently extracts visual features from groups of neighboring grayscale frames. These feature representations are subsequently processed by a 3D convolutional encoder, which captures spatiotemporal information. Generalized mean pooling is then applied to aggregate the resulting features before the final classification stage.

An important aspect of this approach is the separation of spatial and temporal feature extraction, by applying 2D convolutions before 3D convolutions, the architecture first extracts visual information from individual frame groups and subsequently integrates information across time. This design provides an effective balance between spatial representation learning and temporal feature extraction.

Beyond the model architecture, Baikulov's solution employs a multistage training strategy. This includes pretraining on the SoccerNet Action Spotting dataset, transfer learning, and finetuning with longer temporal sequences. The solution also introduces an out of fold prediction based on sampling strategy to increase the probability of selecting challenging training examples. These choices highlight the importance of the training procedure, in addition to the architecture itself, for achieving strong spotting performance.

Another notable approach was proposed by Wang et al. [6], who obtained second place in the 2023 challenge with Boosted Model Ensembling (BME). Their method builds upon the E2E-Spot architecture [7] and focuses on generating and combining multiple candidate models. The candidate models differ in their feature extraction architectures and training configurations. An iterative selection procedure is then used to identify suitable ensemble members and assign their respective weights according to their validation performance.

These two solutions illustrate different strategies for improving ball action spotting: the use of an efficient spatiotemporal architecture and the combination of diverse models through ensemble learning. In particular, Baikulov's slow fusion architecture provides the reference for the approach investigated in this work. We explore an alternative temporal modeling strategy by replacing its 3D convolutional encoder with a Transformer based encoder while retaining the shared 2D visual feature extractor.

C. Convolutional Temporal Modeling

Convolutional neural networks are widely used in video understanding because they can learn hierarchical visual representations directly from image data. A conventional 2D CNN applies convolutional filters across the spatial dimensions of an image, extracting features such as edges, textures, and higher level semantic patterns.

However, video data also contains temporal information that cannot be fully represented by processing frames independently. Three dimensional convolutional neural networks address this limitation by extending convolutional operations to the temporal dimension. Their filters work on both spatial and temporal dimensions, allowing the network to learn local patterns and interactions between neighboring frames.

An alternative to applying 3D convolutions throughout the entire network is the slow fusion strategy. In this approach, spatial features are initially extracted using 2D convolutions and are progressively integrated through following temporal processing stages. This reduces the need to perform full spatio temporal convolutions at the earliest network layers while retaining the ability to learn temporal representations.

The slow fusion architecture employed by Baikulov [5] follows this principle by combining a shared 2D feature extractor with a 3D convolutional temporal encoder. The resulting architecture serves as the convolutional reference for our investigation.

D. Transformer Based Temporal Modeling

Transformers [8] were originally introduced as an attention based architecture for sequence modeling. The central component is the self attention mechanism which enables each element of a sequence to incorporate information from other elements through interactions between query, key, and value representations.

Unlike convolutional operations, which aggregate information through locally defined receptive fields, self attention computes relationships between elements across the input sequence. This allows the model to capture dependencies between temporally separated representations without requiring information to pass exclusively through successive local operations.

Since Transformers do not inherently encode the ordering of input elements, positional information is incorporated into the sequence representations. Positional embeddings provide the model with information about the positions of individual elements, allowing temporal order to influence the attention computation.

In video understanding, this mechanism can be applied to sequences of visual feature representations extracted from individual frames or groups of frames. Then a temporal encoder can learn relationships between these representations and aggregate the resulting information for video level classification.

In this work, we investigate this approach specifically for ball action spotting. In contrast to approaches that globally pool spatial feature maps before temporal modeling, the proposed architecture retains a coarse spatial grid. Temporal attention is first applied independently at each spatial location, followed by spatial attention at each temporal position. This factorized design allows the model to capture both temporal evolution and spatial relationships. This provides an alternative to the convolutional temporal modeling strategy employed by the reference architecture.

III. Proposed Method

A. Architecture Overview

The proposed model is a single stage architecture designed to detect ball related actions by exploiting spatial appearance and temporal information. It combines a 2D convolutional backbone with a factorized spatial temporal Transformer. The backbone extracts visual representations from groups of sampled grayscale frames, while the Transformer models temporal dependencies and spatial relationships in the resulting feature maps.

Each input segment contains 15 grayscale frames sampled at a stride of two frames in the original video. The sampled sequence is divided into five non overlapping groups of three sampled frames. Each group is processed by a shared 2D backbone. The resulting feature maps are projected to 192 channels and arranged as a spatiotemporal feature tensor.

Rather than collapsing each feature map into a single vector before temporal modeling, the proposed architecture preserves a coarse spatial representation throughout the attention stages. Temporal attention is first applied independently at each spatial location. Spatial attention then enables information exchange between locations within each temporal step. Finally, the encoded features are averaged over temporal and spatial positions and passed to a linear classifier that predicts pass and drive.

B. Frame Grouping and 2D Feature Extraction

Let the input video segment be represented by

$$X \in \mathbb{R}^{B\times T\times H\times W},$$

where B is the batch size, $T = 15$ is the number of sampled frames, and H and W denote frame height and width. The 15 sampled frames are grouped into five stacks of three frames, giving

$$X_s \in \mathbb{R}^{(B\cdot 5)\times 3\times H\times W}.$$

Each stack is treated as a three channel input and processed by the same EfficientNetV2-B0 backbone, initialized with ImageNet-1K pretrained weights [9]. The selected backbone feature stage produces 192 channel feature maps with a spatial resolution of 23×40 for the input configuration used in this work. A 1×1 convolution, batch normalization, and SiLU activation project the features to 192 channels. The projected maps are reshaped to recover the temporal dimension:

$$F \in \mathbb{R}^{B\times 5\times 192\times 23\times 40}.$$

C. Spatial Feature Reduction

Applying attention directly at the full feature map resolution would produce a large number of tokens. Adaptive average pooling is therefore applied independently to each feature map, reducing its spatial dimensions from 23×40 to 6×10:

$$F_p \in \mathbb{R}^{B\times 5\times 192\times 6\times 10}.$$

The resulting grid contains 60 spatial locations per temporal position, each represented by a 192 dimensional feature vector. This operation reduces the token count without collapsing the complete feature map into a single vector; the coarse spatial layout is retained for the subsequent attention operations.

D. Factorized Spatial Temporal Transformer

The Transformer consists of two sequential stages: temporal attention and spatial attention. The embedding dimension is 192 throughout both stages.

The feature tensor is rearranged so that each spatial location is associated with a sequence of five feature vectors, one for each temporal group:

$$F_t \in \mathbb{R}^{(B\cdot 60)\times 5\times 192}.$$

Each spatial location is therefore processed as an independent temporal sequence. A learnable temporal positional embedding is added to encode the ordering of the five temporal positions. The temporal encoder contains two Transformer layers, each with eight attention heads, a feed forward dimension of 768, and dropout of 0.1. Temporal attention models dependencies between the five frame groups independently at each of the 60 spatial locations.

After temporal encoding, the tensor is reshaped to

$$F'_t \in \mathbb{R}^{B\times 5\times 60\times 192}.$$

For each temporal position, the 60 spatial feature vectors form a sequence:

$$F_s \in \mathbb{R}^{(B\cdot 5)\times 60\times 192}.$$

A learnable spatial positional embedding represents the locations in the 6×10 grid. The spatial encoder contains one Transformer layer with eight attention heads, a feed forward dimension of 768, and dropout of 0.1. This stage enables information exchange between spatial locations independently at each temporal position.

E. Feature Aggregation and Classification

Following the spatial Transformer, the encoded representation has the shape

$$F_{\text{out}} \in \mathbb{R}^{B\times 5\times 60\times 192}.$$

The temporal and spatial dimensions are averaged to obtain one feature vector per input segment:

$$z_b = \frac{1}{5 \cdot 60} \sum_{t=1}^{5} \sum_{s=1}^{60} F_{\text{out}}[b, t, s, :].$$

The resulting vector has 192 dimensions. A fully connected layer maps it to two output scores corresponding to pass and drive:

$$\hat{y}_b = W z_b + b.$$

IV. Experimental Setup

A. Dataset and Task

The experiments are conducted in the context of the SoccerNet Ball Action Spotting Challenge 2023 [4]. The task consists of identifying two types of ball related actions: passes and drives. A pass corresponds to the moment when the ball leaves a player's foot, whereas a drive corresponds to the instance when a player controls the ball.

The dataset consist of broadcast soccer videos annotated with temporal action labels. The objective is to predict the temporal locations of the annotated actions from the video frames.

B. Input Representation

The model receives 15 grayscale frames sampled with a temporal stride of two frames from the original video. The sampled frames are divided into five non overlapping groups of three frames. The input resolution is 1280×736. The frame processor is cofigured to apply padding and normalization operation. Both padding and filling value are zero. The resulting groups are the three channel inputs to the shared 2D backbone.

C. Training Configuration

The proposed model was trained following the training procedure of the original Ball Action Spotting Challenge solution by Baikulov [5]. The original training configuration was retained, including the optimizer, learning rate schedule, number of epochs, and loss function.

The only modification concerned the batch size, which was reduced from 2 to 1 due to GPU memory constraints. While the original implementation was trained on an NVIDIA GeForce RTX 3090 Ti, our experiments were conducted on an NVIDIA GeForce RTX 3080. This adjustment allowed the model to be trained within the available hardware resources while preserving the remaining training settings.

The experimental evaluation considers the model obtained directly after the training procedure. No additional finetuning stages, training phases, or post processing operations were applied to the predictions.

The model was trained using the configuration adapted from Baikulov's reference implementation [5]. The input videos were represented by sequences of 15 grayscale frames, sampled with a temporal step of two frames and grouped into five stacks of three consecutive frames. Each frame was resized and padded to a resolution of 1280 × 736.

The model was optimized using AdamW, with a base learning rate of 3×10^{-4} and a batch size of 1. Training employed mixed precision computation and an epoch size of 6,000 samples. The training schedule consisted of two stages: a six epoch warmup stage followed by a 30 epoch training stage. Focal loss was used, with a focusing parameter $\gamma = 1.2$.

An exponential moving average of the model parameters was maintained with a decay factor of 0.999. The EfficientNetV2-B0 backbone was initialized with pretrained ImageNet-1K weights and was not frozen during training. The batch size was reduced from the reference configuration to accommodate the available GPU memory.

D. Dataset Partitioning and Training Protocol

We adopted the seven fold cross-validation strategy of Baikulov [5]. Each labelled game was assigned to one fold. For each fold, six folds were used for training, while the remaining fold was reserved for validation. This procedure was repeated seven times, resulting in seven fold specific models.

The proposed Transformer architecture was trained using this protocol. The original training configuration was retained, with the exception of the batch size, which was reduced from 2 to 1 to accommodate GPU memory constraints. No additional finetuning or postprocessing was performed.

E. Evaluation Metrics

Performance is evaluated using mean Average Precision at a temporal tolerance of one second (mAP@1), following the official SoccerNet Ball Action Spotting Challenge 2023 evaluation protocol. Average Precision is computed independently for each event category, Pass and Drive, and mAP@1 is obtained by averaging the two class specific AP values. The one second tolerance reflects the high temporal precision required to localize ball related events.

V. Results and Discussion

A. Cross-Validation Results

The proposed Transformer based architecture was evaluated using the seven fold cross-validation protocol. Each fold uses six games for training and one held out game for validation. The evaluation script computes mAP@1 independently for each held out game.

The model achieves a mean cross-validation mAP@1 of **77.45%**, with a standard deviation of **5.47 percentage points** across the seven folds. Individual fold results range from

TABLE I
COMPARISON WITH THE RESULTS REPORTED FOR BAIKULOV'S CNN BASED SOLUTION. THE TWO BAIKULOV ROWS CORRESPOND TO DIFFERENT TRAINING AND INFERENCE CONFIGURATIONS.

Method	CV mAP@1 (%)	Public test mAP@1(%)
Baikulov basic training	79.06	84.26
Baikulov final model	80.49	87.04
Proposed Transformer	77.45	81.79

69.97% to 83.25%, indicating some variation in performance across games.

For reference, Baikulov reports a cross-validation mAP@1 of **79.06%** for the original CNN based solution. The proposed Transformer therefore obtains a result 1.61 percentage points lower than the reported reference. This comparison provides an indication of the performance difference, although it should not be interpreted as a fully controlled architectural ablation.

B. Public Test Results

The public test evaluation follows the same out of fold prediction procedure as Baikulov's implementation. Predictions are generated using the two fold specific models that include the public test games, allowing the results to be compared under the same evaluation protocol.

The original CNN based solution reports **84.26%** mAP@1 on the public test set. The proposed Transformer achieves **81.79%**mAP@1.

C. Comparison with the Reference Solution

Table I summarizes the cross-validation and public test results. Baikulov's basic training and final model rows refer to different configurations and must not be interpreted as results from one unchanged model. The basic training configuration was reported at 79.06% CV and 84.26% public test mAP@1, while the final model was reported at 80.49% CV and 87.04% public test mAP@1 [5].

VI. CONCLUSION

The experiments show that factorized spatial temporal attention is a viable alternative for modeling video features in ball action spotting, although the current results remain below those reported for the reference CNN based solution. Since the two approaches were not evaluated under identical training and inference conditions, further controlled experiments are necessary to determine the contribution of the proposed attention mechanism. In particular, comparing both architectures using the same training schedule and prediction pipeline would provide a more reliable assessment of their relative performance.

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